



Data Warehousing/Mining

Introduction

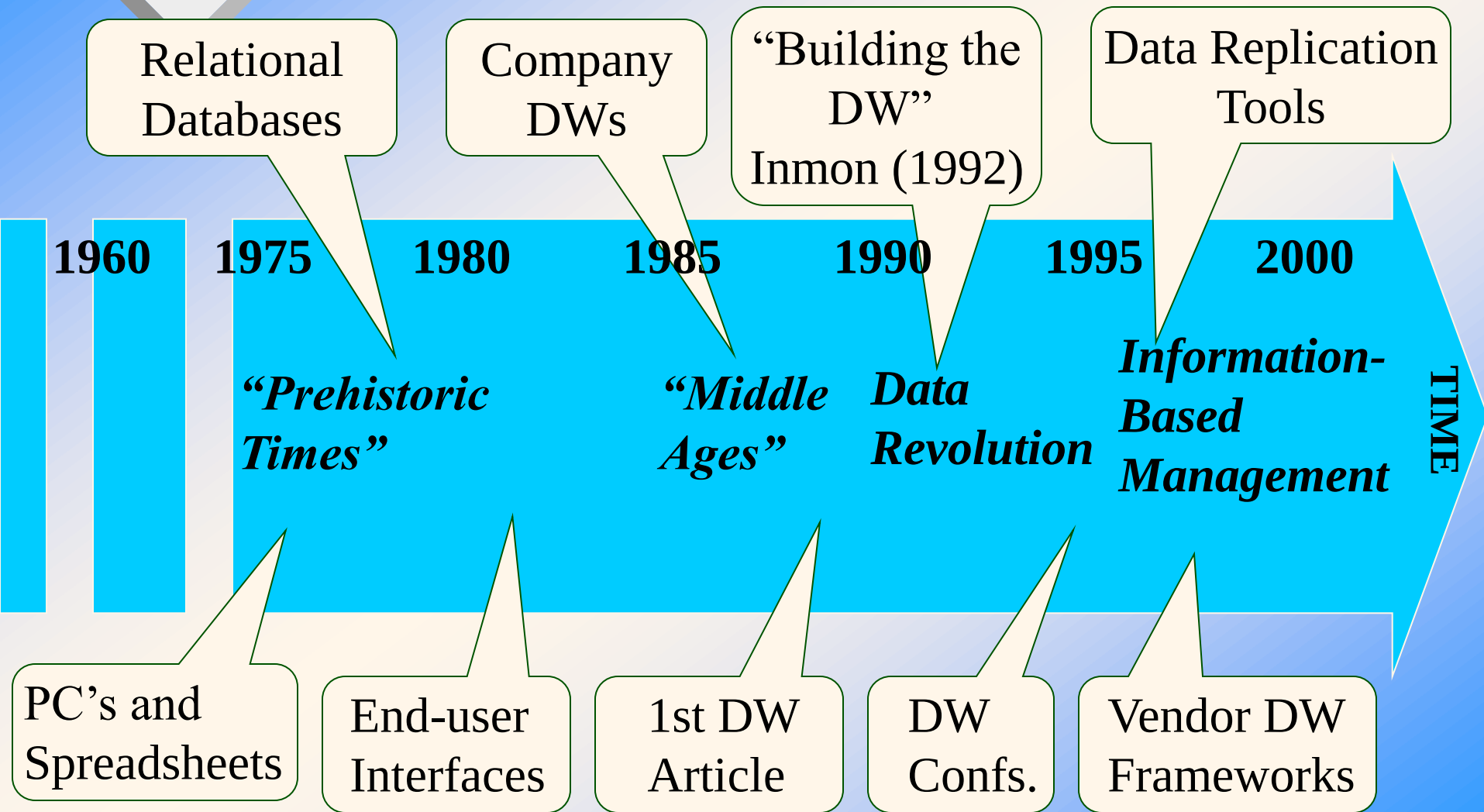




Outline of Lecture

- Brief History of Data Warehousing
- What is a Data Warehouse?
- Need For Strategic Information
- Information Crisis
- Operational and Decision Support System
- Difference B/W standard DB and Data warehouse

Data Warehouse Evolution





Escalating Need For Strategic Information

Organizations need information to formulate the business strategies, establish Goals, set Objectives

e.g.

- Increase the customer by 10% over the next 5 years
- Gain market share by 15% in the next 2 years
- Increase product quality levels in the top five product groups



The Information Crisis

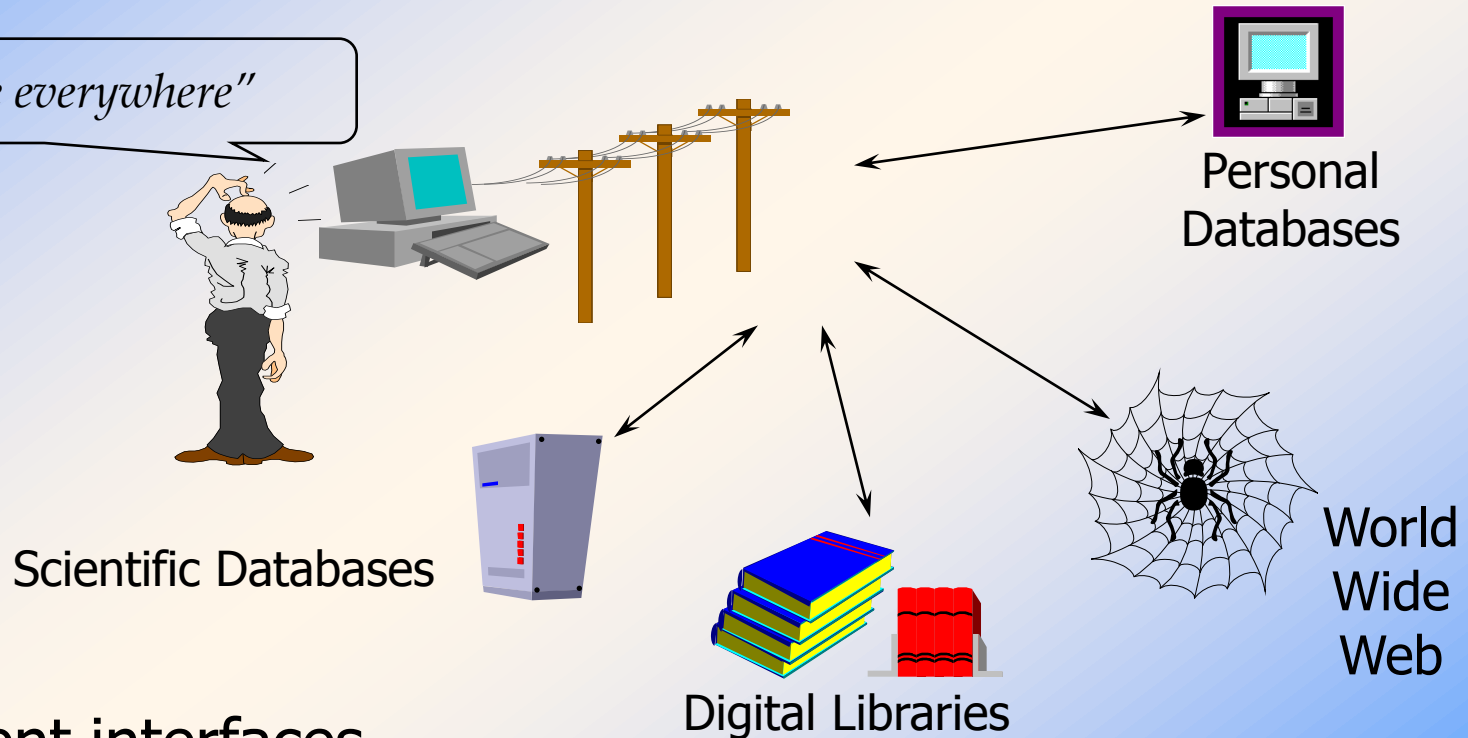
- Information is said to be doubled every 18 months
- Organizations have tons of data available

Then why information Crisis?

Why cant organizations convert the data into useful information for strategic decision making?

Problem: Heterogeneous Information Sources

"Heterogeneities are everywhere"

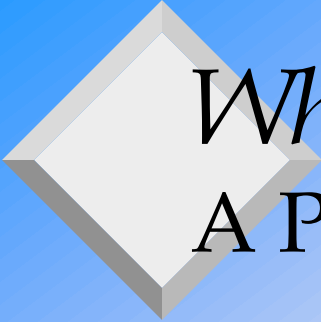


- Different interfaces
- Different data representations
- Diverse structure of databases
- Duplicate and inconsistent information



About Some Definitions

- What is data?
- What is information?
- What is Warehouse?



What is a Data Warehouse?

A Practitioners Viewpoint

“A data warehouse is simply a single, complete, and consistent store of data obtained from a variety of sources and made available to end users in a way they can understand and use it in a business context.”

-- Barry Devlin, *IBM Consultant*



A Data Warehouse is...

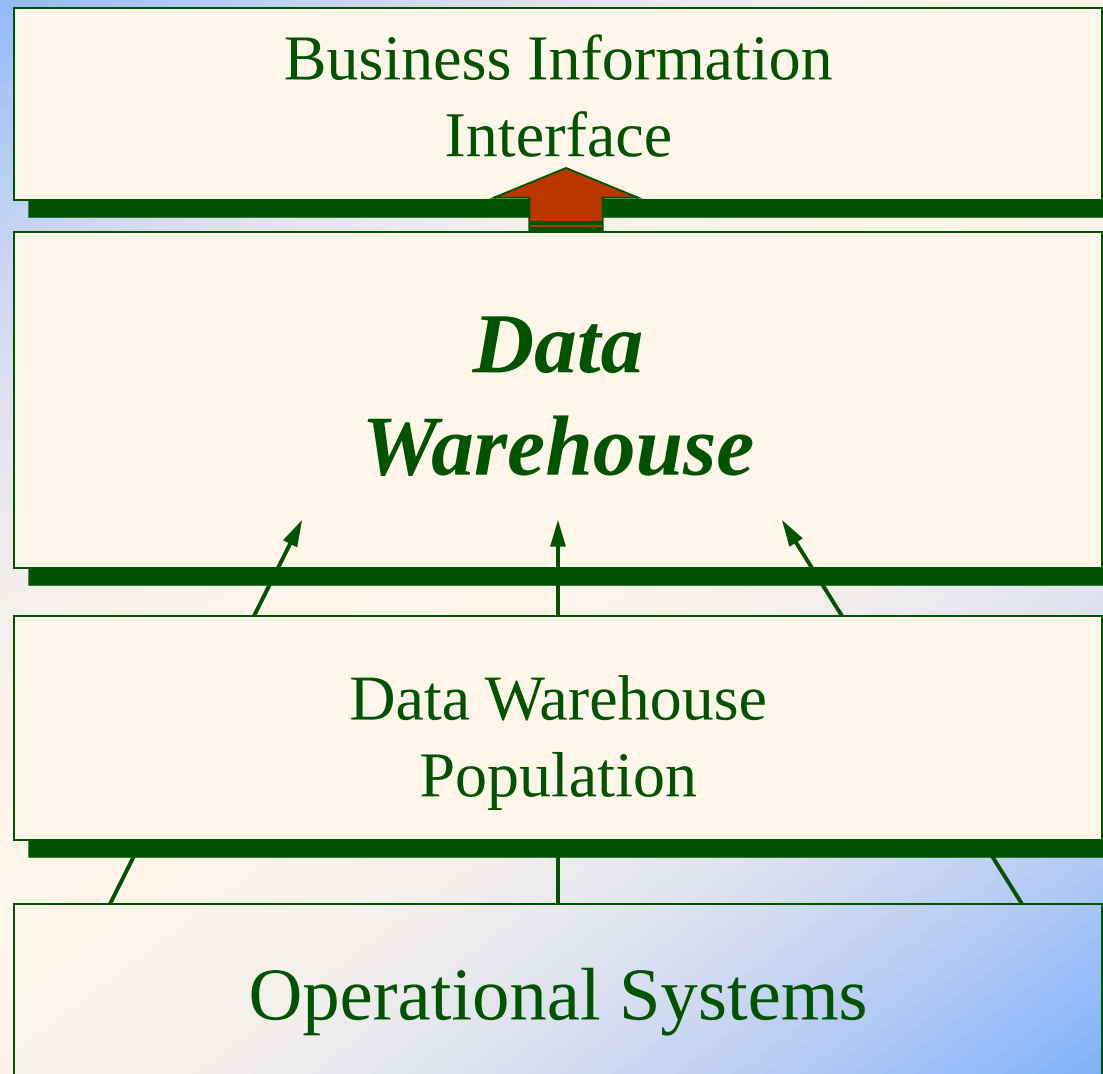
- Stored collection of diverse data
 - A solution to data integration problem
 - Single repository of information
- Subject-oriented
 - Organized by subject, not by application
 - Used for analysis, data mining, etc.
- Large volume of data (Gb, Tb)
- Non-volatile
 - Historical
 - Time attributes are important



A Data Warehouse is... (continued)

- Updates infrequent
- Examples
 - All transactions EVER at WalMart
 - Complete client histories at insurance firm
 - Stockbroker financial information and portfolios

Summary



What is Operational and Decision Support System

Operational Systems

- Making the wheels of Business Turn
 - Take an order
 - Process a claim
 - Make shipment
 - Generate an invoice
 - Receive cash
 - Reserve an airline seat



What is Operational and Decision Support System (Contd...)



Decision Support System

- Watching the wheels of business turn
 - Show the top selling products
 - Show the problem regions
 - Tell me why (drill down)
 - Let me see other data (drill across)
 - Alert me when a district sells below target

Difference

	Operational	Informational
Data Content	Current Values	Archived, derived, optimized
Data Structure	Optimized for transaction	Optimized for complex queries
Access Frequency	High	Medium to Low
Access Type	Read, update, delete	Read
Usage	Predictable, repetitive	Ad hoc, random, Heuristic
Response Time	Sub seconds	Several Seconds to Minutes
Users	Large Number	Relatively Small number



Warehouse is a Specialized DB

Standard DB

- Mostly updates
- Many small transactions
- Mb - Gb of data
- Current snapshot
- Index/hash on p.k.
- Raw data
- Thousands of users (e.g., clerical users)

Warehouse

- Mostly reads
- Queries are long and complex
- Gb - Tb of data
- History
- Lots of scans
- Summarized, reconciled data
- Hundreds of users (e.g., decision-makers, analysts)



Warehousing and Industry

- Warehousing is big business
 - \$2 billion in 1995
 - \$3.5 billion in early 1997
 - About \$8 billion in 1998 [Metagroup]
- WalMart has largest warehouse
 - 900-CPU, 2,700 disk, 23 TB Teradata system
 - ~7TB in warehouse
 - 40-50GB per day



Data Warehousing: Two Distinct Issues

(1) How to get information into warehouse

“Data warehousing”

(2) What to do with data once it's in warehouse

“Warehouse DBMS”

- Both rich research areas
- Industry has focused on (2)



Thank You Very Much